

CONTACT INFORMATION	Web: https://i.dk-lab.xyz/ E-mail: dxu27@ncsu.edu GIC Lab: [Link]	Google Scholar: [Link] LinkedIn: [Link] X: [Link] Updated: Sep. 8, 2026
RESEARCH AGENDA	My research studies learning in open-ended environments—settings in which relevant tasks, hypotheses, solution strategies, and operating conditions can expand or change through interaction. I ask how learning systems should decide <i>what is worth exploring or computing</i>, <i>what the available evidence supports under uncertainty</i>, and <i>what internal models or capabilities should change as experience accumulates</i>. These questions define an exploration–evidence–revision loop connecting efficient exploration and computation, prediction and decision under uncertainty, and continual learning and model evolution under bounded computation and verification. I study this loop in scientific, physical, human, and computing environments, including autonomous experimentation, multi-robot systems, education and creative computing, and networked systems. I lead the Generative Intelligent Computing (GIC) Lab at NC State.	
ACADEMIC APPOINTMENT	Assistant Professor, North Carolina State University • Department of Computer Science	Aug. 2022–Present
EDUCATION	Ph.D., Penn State University, PA, USA M.S., University of Chinese Academy of Sciences, Beijing, China B.E., Renmin University of China, Beijing, China	2022 2017 2014
RESEARCH HIGHLIGHTS	• Scholarship. 80+ peer-reviewed publications; 7,049 citations (h-index: 28; i10-index: 53; Google Scholar, Sep. 8, 2026); 10 issued U.S. patents across 6 invention families. • Funding. PI/Co-PI on five active NSF-supported projects; four cash awards total approximately \$3.43M in collaborative project funding, plus an NSF NAIRR compute award. Additional research support from Microsoft and NVIDIA. • Recognition. NC State College of Engineering Dean’s Applied AI Research Accelerator Award (2026; \$100K, PI); AAAI Best Technical Demonstration Award (2025); AAAI New Faculty Highlights (2025). • Paper distinctions. ICLR’26 Oral (top 5%); AAAI’25 Oral (top 5%); CVPR’23 Highlight (top 2.5%); IEEE IPCCC’24 Best Paper Runner-Up; ICCCN’23 Best Paper.	
RESEARCH CONTRIBUTIONS	• Efficient Exploration & Computation. Study adaptive rather than uniform computation: where to allocate model capacity, data, and inference-time reasoning when candidate computations differ in cost and information value. <i>Representative:</i> Adaptix (adaptive LLM decoding), AAAI’25 Oral [Paper] [Code]. • Prediction & Decision under Uncertainty. Study what evidence justifies a prediction or action under distribution shift, partial observation, or model uncertainty, and when additional information or verification is warranted. <i>Representative:</i> DALD (black-box LLM-generated-text detection), NeurIPS’24 [Paper] [Code]. • Continual Learning & Model Evolution. Study stability–plasticity under constrained capacity: what should remain stable, what should change, and how new capabilities can be incorporated without uncontrolled forgetting or unbounded growth. <i>Representative system:</i> Gentopia (composable specialized LLM agents), EMNLP’23 System Demo [Paper] [Code].	
RESEARCH FUNDING	Active Federal Awards • NSF NAIRR Pilot	2025–2026
	<i>Advancing Retrieval-Augmented AI to Deliver Transparent, Trustworthy Insights from Massive Ocean Datasets</i>	

- PI; Co-PIs: Paul Liu and Ruoying He; **\$30,000 H100 GPU compute allocation.**
- **NSF CAIG** 2026–2029
Physics-Informed Deep Learning for Understanding Coastal Hypoxia Formation Mechanisms
Co-PI (AI); Lead PI: Paul Liu; team: Ruoying He (NC State) and Kehui Xu (LSU); **\$1,049,875 collaborative total; \$800,013 to NC State.**
 - **NSF ExLENT (Beginnings)** 2026–2028
Experiential Learning for In-Service Teachers: Augmenting Teaching and Learning with Generative AI
Co-PI (AI); PI: Tiffany Barnes; team: Joey Huang, Xiaoyi Tian, and Andrew Grillo-Hill (WestEd); **\$1,000,000.**
 - **NSF CyberTraining** 2024–2027
EcoTern: Pioneering a CI Workforce for Sustainable and Transdisciplinary Environmental Science Research
Co-PI (AI); NC State: Xipeng Shen (PI), Ruoying He; FIU: Wenqian Dong (PI), Jayantha Obeysekera, Jason Liu; **\$979,901 collaborative total; \$439,902 to NC State.**
 - **NSF HEGS** 2024–2027
Harnessing Historical Microdata and Street-View Images to Assess Transit-Induced Neighborhood Change
Co-PI (AI); PI: Eleni Bardaka; **\$396,600.**

Selected External & Institutional Research Support

- **NC State College of Engineering Dean’s Applied AI Research Accelerator** 2026–2027
Transforming Nanoscale Sensing with Scalable Agentic AI and Multi-Functional Microscopy
PI / Project Lead; Co-PIs: Yang Zhang and Caroline Laplante; **\$100,000.**
- **Microsoft Accelerating Foundation Models Research** 2024–2025
Scalable and Adaptable Evaluation of LLM Trustworthiness
PI; **\$50,000 in Azure credits and gift funds.**
- **NVIDIA Academic Grant Program** 2025
Orchestrating Multi-Level Network Modeling and Scientific Simulation with LLMs
Co-PI; PI: Yuchen Liu; **\$20,000 in A100 GPU compute.**
- **Additional NC State research support:** PI-SEAS Interdisciplinary Faculty Research Award (\$1.5K, PI, 2026; autonomous materials experimentation with Martin Seifrid, MSE); CMI Ideation Award (\$30K, Co-PI, 2025–2026); Data Science and AI Academy Seed Grant (\$30K, Co-PI, 2024–2025); AI Innovation Summer Project (\$7.5K, Co-PI, 2026); DELTA Grant (\$8K, Co-PI, 2025–2026); TELS Grant (\$1.6K, Co-PI, 2025).

RESEARCH LEADERSHIP

- **Organizer**, NC State Computer Science AI Seminar Series 2026–Present [\[Link\]](#).
- **Chair / Organizer**, Workshop on DL–Hardware Co-Design for AI and Generative AI Acceleration: AAAI’23; DAC’24, DAC’25, DAC’26.
- **Co-Chair**, Resource-Efficient Learning for Knowledge Discovery (RelKD): KDD’23, KDD’24, KDD’25, KDD’26.
- **Co-Chair**, Dataset Distillation for Computer Vision Workshop: CVPR’24.
- **Invited Chair**, New Faculty Highlights: IEEE Big Data’26.
- **NSF Panelist / Reviewer** 2023–2026
- **Area Chair / Senior Program Committee:** LREC-COLING’24; AAAI’24–’27; IJCAI’21.
- **Column Editor**, ACM SIGAI AI Matters.

RESEARCH SYSTEMS

- **Gentopia** — *Project Lead*. Open-source, config-driven framework for constructing, evaluating, composing, and sharing specialized LLM agents with configurable tools and memory. EMNLP’23 System Demo; 300+ GitHub stars. [\[System\]](#) [\[Paper\]](#).
- **SIMBA** — *AI / Agentic Workflow Lead*. Agentic AI platform integrating natural-language interaction, workflow orchestration, localization, spectral analysis, and deep-learning-based denoising for single-molecule multidimensional imaging. bioRxiv’26. [\[System\]](#) [\[Preprint\]](#).

- **OceanAI** — *NSF NAIRR Pilot PI for platform scaling*. Retrieval-grounded platform connecting LLM interaction with authoritative oceanographic data for transparent analysis, visualization, and provenance-aware scientific responses. WACV’26 Demo. [System] [Paper].
- **MerryQuery** — *Faculty Co-Lead / Co-PI*. Course-aligned educational AI assistant combining retrieval grounding, source citation, controllable assistance, and instructor analytics. AAAI’25 Best Technical Demonstration Award; deployed in a 250+ student course. [System].
- **ArtAlgo** — *Co-PI / AI Collaborator*. Creative-computing platform combining visual coding with AI-assisted design to study computational thinking, creativity, and human–AI collaboration. ISLS’26. [System].

EDUCATION & OUTREACH

- **Wolf AI Summer Bootcamp, NSF EcoTern** 2026
Led NC State activities on modern AI for scientific and engineering applications, including machine learning, LLMs, agentic AI, knowledge graphs, multimodal learning, and AI-enabled scientific workflows.
- **AI Training Workshop for Water Resources and Environmental Applications, EWRI’26** 2026
Co-organized hands-on AI training for researchers and practitioners in environmental and water-resources engineering.
- **Integrating ChatGPT into K–12 Classrooms** 2023–2024
Co-organized a three-workshop series for K–12 educators on generative AI for teaching and learning; resulting research was published at IEEE FIE’24.
- **Research Mentoring** 2023–Present
Mentored undergraduate and graduate researchers through the GIC Lab and NC State programs including COE REU, GCSP-REU, GEARS, NSF SRCA-REU, and CSC 298.
- **Selected Mentoring Outcomes.** Aditya Basarkar continued from GIC Lab undergraduate research into the Ph.D. program and was the sole departmental recipient of the 2026 Tony McLean Brown Graduate Scholarship. Four undergraduate mentees received six NC State COE REU Awards during 2023–2026, including two to Ethan Xu in 2026. Additional recognition includes the AAAI/SIGAI’25 Doctoral Consortium (Benyamin T. Tabarsi, co-advised), AAAI’25 Undergraduate Consortium and NCUR’25 (Precious Donkor), and CS Graduate Merit Awards (Travis Thompson and Stephen Liu, 2026).

PEER-REVIEWED CONFERENCE AND JOURNAL PAPERS

- [1] D. Wu, J. Liu, X. Lin, **D. Xu**, P. Gao, and Y. Liu. Revisiting Multi-Level Network Modeling and Simulation With Domain-Adapted LLMs [J]. **IEEE Transactions on Network Science and Engineering**, 2026.
- [2] A. Zhou, K. Wang, Y. Liu, **D. Xu**, and L. Xiang. Multi-task Spatio-temporal Modeling: Towards Accurate and Robust Plant Growth Prediction in the Real World [J]. **Smart Agricultural Technology**, 2026.
- [3] J. Liu, S. Tang, J. Cui, **D. Xu**, and Z. Shen. BiGain: Unified Token Compression for Joint Generation and Classification [C]. The IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR’26**).
- [4] J. Huang, C. Varda, and **D. Xu**. Examining AI as Design Partner in Creative Computing: Digital Dialogue to Embodied Expression [C]. The 20th International Conference of the Learning Sciences (**ICLS’26**).
- [5] H. Li, **D. Xu**, M. Chen, and Y. Liu. Agentic Open RAN: A Deterministic and Auditable Framework for Intent-Driven Radio Control [C]. IEEE International Conference on Communications (**ICC’26**).
- [6] J. Yao, S. Duan, X. Yi, **D. Xu**, P. Zhang, T. Lu, N. Gu, Z. Dou, and X. Xie. AdaEM: An Adaptively and Automated Extensible Measurement of LLMs’ Value Difference [C].

The International Conference on Learning Representations (**ICLR'26, Oral Presentation; top 5%**).

- [7] H. Darabipourshiraz, D. Smyslova, **D. Xu**, S. Jiang, and D. Long. BiasViz: A Project-Based, Narrative-Centered Learning Tool for Engaging Middle School Students in Critical Thinking about AI Biases [C]. The ACM CHI Conference on Human Factors in Computing Systems (**CHI'26**).
- [8] P. Gu, H. Li, H. Tang, **D. Xu**, E. Enriquez, D. Kim, B. Fu, and D. Z. Chen. Integrating Multi-scale and Multi-filtration Topological Features for Medical Image Classification [C]. The IEEE/CVF Winter Conference on Applications of Computer Vision (**WACV'26**).
- [9] H. Mao, O. K. Venkataraman, **D. Xu**, K.-H. Song, and Y. Zhang. A Monte Carlo-based Deep Learning Framework for Accurate Single-Molecule Spectroscopic Imaging [C]. **Proceedings of SPIE, Single Molecule Spectroscopy and Superresolution Imaging XIX, Vol. 13858, 2026**.
- [10] H. Mao, Y. Liu, O. K. Venkataraman, M. A. Shahid, C. Laplante, **D. Xu**, K.-H. Song, and Y. Zhang. Framework for Accurate Single-Molecule Spectroscopic Imaging Analyses Using Monte Carlo Simulation and Deep Learning [J]. **Analytical Chemistry, 97(30):16250–16258, 2025**.
- [11] K. Zhao, Y. Du, R. Zhang, L. Zhan, **D. Xu**, P. Gu, and H. Tang. Who Matters More in Radiology Report Generation: Vision Encoders or Language Models? [C]. IEEE International Conference on Data Mining Workshops, Synergy of AI and Multimodal Biomedical Data Mining Workshop (**ICDMW'25, SAIMBio**).
- [12] S. Dai, L. Li, K. Zhao, E. Lila, P. K. Crane, H. Huang, **D. Xu**, H. Tang, and L. Zhan. Why Text Prevails: Vision May Undermine Multimodal Medical Decision Making [C]. IEEE International Conference on Data Mining Workshops, Synergy of AI and Multimodal Biomedical Data Mining Workshop (**ICDMW'25, SAIMBio**).
- [13] H. Huang, **D. Xu**, H. Zhang, P. Gao. Non-Overlap-Aware Egocentric Pose Estimation for Collaborative Perception in Connected Autonomy [C]. The 2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (**IROS'25 (Findings)**).
- [14] C. Shen, Z. Chen, D. Luo, **D. Xu**, H. Chen, and J. Ni. Exploring Multi-Modal Data with Tool-Augmented LLM Agents for Precise Causal Discovery [C]. **Findings of the Association for Computational Linguistics: ACL 2025**.
- [15] Z. Kong, **D. Xu**, Z. Li, P. Dong, H. Tang, Y. Wang, S. Mukherjee. AutoViT: Achieving Real-Time Vision Transformers on Mobile via Latency-aware Coarse-to-Fine Search [J]. International Journal of Computer Vision
- [16] Z. Peng, Y. Liu, G. Li, Z. Yang, M. Chen, **D. Xu**, X. Lin. Generative Artificial Intelligence Models for Emerging Communication Systems: Fundamentals and Challenges [J]. IEEE Communications Magazine
- [17] C. Egersdoerfer, A. Sareen, J. L. Bez, S. Byna, **D. Xu**, D. Dai. Augmenting LLMs for HPC I/O Performance Diagnosis [C]. The 37th International Conference on Scalable Scientific Data Management (**SSDBM'25**)
- [18] B. Tabarsi, A. Basarkar, X. Liu, **D. Xu**, T. Barnes. MerryQuery: A Trustworthy LLM-Powered Tool Providing Personalized Support for Educators and Students [C]. The 39th Annual AAAI Conference on Artificial Intelligence (Demo Track) (**AAAI'25 (Demo Track)**)

- [19] X. Liu, B. Lei, R. Zhang, and **D. Xu**. Adaptive Draft-Verification for Efficient Large Language Model Decoding [C]. The 39th AAAI Conference on Artificial Intelligence (**AAAI'25, Oral Presentation**).
- [20] X. Liu, H. Lv, F. Ma, C. Wang, **D. Xu**. DyESP: Accelerating Hyperparameter-Architecture Search via Dynamic Exploration and Space Pruning [C]. The 39th Annual AAAI Conference on Artificial Intelligence Spring Symposium (**AAAI Spring Symposium'25**)
- [21] C. Egersdoerfer, A. Sareen, J. Bez, S. Byna, **D. Xu**, D. Dai. IOAgent: Democratizing Trustworthy HPC I/O Performance Diagnosis Capability via LLMs [C]. The 39th IEEE International Parallel & Distributed Processing Symposium (**IPDPS'25**)
- [22] D. Luo, T. Zhao, W. Cheng, **D. Xu**, F. Han, W. Yu, X. Liu, H. Chen, and X. Zhang. Towards Inductive and Efficient Explanations for Graph Neural Networks [J]. **IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)**, 46(8):5245–5259, 2024.
- [23] C. Zeng, S. Tang, X. Yang, Y. Chen, Y. Sun, Z. Xu, Y. Li, H. Chen, W. Cheng, **D. Xu**. Improving Logits-based Detector without Logits from Black-box LLMs [C]. The 38th Annual Conference on Neural Information Processing Systems (**NeurIPS'24**)
- [24] H. Reichert, B. Tabarsi, Z. Zhang, C. Fennell, I. Bhandari, D. Robinson, M. Drayton, C. Crofton, M. Lococo, **D. Xu**, T. Barnes. Empowering Secondary School Teachers: Creating, Executing, and Evaluating a Transformative Professional Development Course on ChatGPT [C]. IEEE Frontiers in Education Conference 2024 (**FIE'24**)
- [25] S. Tang, Y. Wang, C. Ding, Y. Liang, Y. Li, and **D. Xu**. AdaDiff: Accelerating Diffusion Models through Step-Wise Adaptive Computation [C]. The 18th European Conference on Computer Vision (**ECCV'24**)
- [26] J. Liu, Z. Peng, **D. Xu**, Y. Liu. Revolutionizing Wireless Modeling and Simulation with Network-Oriented LLMs [C]. The 43rd IEEE International Performance Computing and Communications Conference (**IPCCC'24**)
- [27] Z. Zhang, Y. Liu, Z. Peng, M. Chen, **D. Xu**, and S. Cui. Digital Twin-Assisted Data-Driven Optimization for Reliable Edge Caching in Wireless Networks [J]. IEEE Journal on Selected Areas in Communications (**IEEE JSAC**, 2024, **Impact Factor is 16.4**)
- [28] P. Dong, J. Zhuang, Z. Yang, S. Ji, Y. Li, **D. Xu**, H. Huang, J. Hu, A. Jones, Y. Shi, Y. Wang, P. Zhou. EQ-ViT: Algorithm-Hardware Co-Design for End-to-End Acceleration of Real-Time Vision Transformer Inference on Versal ACAP Architecture [C]. The International Conference on Hardware/Software Codesign and System Synthesis (**CODES+ISSS'24**)
- [29] X. Wang, S. Duan, X. Yi, J. Yao, S. Zhou, Z. Wei, P. Zhang, **D. Xu**, M. Sun, X. Xie. On the Essence and Prospect: An Investigation of Alignment Approaches for Big Models [C]. International Joint Conference on Artificial Intelligence (Survey Track) (**IJCAI'24**)
- [30] Y. Wang, Q. Zhao, **D. Xu**, and X. Liu. Purpose Enhanced Reasoning through Iterative Prompting: Uncover Latent Robustness of ChatGPT on Code Comprehension [C]. 2024 International Joint Conference on Artificial Intelligence (**IJCAI'24**)
- [31] X. Luo, Z. Li, Z. Peng, **D. Xu**, Y. Liu. RM-Gen: Conditional Diffusion Model-Based Radio Map Generation for Wireless Networks [C]. International Federation for Information Processing Networking Conference (**IFIP/IEEE Networking'24**)
- [32] B. Lei, **D. Xu**, R. Zhang, and B.K Mallick. Embracing Unknown Step by Step: Towards Reliable Sparse Training in Real World [J]. (**Transactions on Machine Learning Research**, 2024)

- [33] B. Lei, **D. Xu**, R. Zhang, S. He, B. K. Mallick. Balance is Essence: Accelerating Sparse Training via Adaptive Gradient Correction [C]. The 2024 Conference on Parsimony and Learning (**CPAL'24**)
- [34] Z. Zhang*, Z. Dong*, Y. Shi, N. Matsuda, T. Price, **D. Xu**. Students' Perceptions and Preferences of Generative Artificial Intelligence Feedback for Programming [C]. The 14th Symposium on Educational Advances in Artificial Intelligence (**AAAI/EAAI'24**)
- [35] Z. Wang, Q. Zhao, J. Cui, X. Liu, and **D. Xu**. AutoST: High-performance and Energy-efficient Spiking Transformer Architecture Search [C]. The 2024 IEEE International Conference on Acoustics, Speech, and Signal Processing (**ICASSP'24**)
- [36] L. Zhang, J. Zhang, B. Lei, S. Mukherjee, X. Pan, B. Zhao, C. Ding, Y. Li, and **D. Xu**. Accelerating Dataset Distillation via Model Augmentation [C]. The 34th IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR'23, Highlight Paper**)
Acceptance rate: **235/9155=2.5%**
- [37] S. Tang, Y. Wang, Z. Kong, T. Zhang, Y. Li, C. Ding, Y. Wang, Y. Liang, and **D. Xu**. You Need Multiple Exiting: Dynamic Early Exiting for Accelerating Unified Vision Language Model [C]. The 34th IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR'23**)
- [38] B. Xu, X. Liu, H. Shen, Z. Han, Y. Li, M. Yue, Z. Peng, Y. Liu, Z. Yao, and **D. Xu**. Gen-topia: A Collaborative Platform for Tool-Augmented LLMs [C]. The 2023 Conference on Empirical Methods in Natural Language Processing. (**EMNLP'23, System Track**)
- [39] J. Li, Q. Lei, W. Cheng, and **D. Xu**. Towards Robust Pruning: An Adaptive Knowledge-Retention Pruning Strategy for Language Models [C]. The 2023 Conference on Empirical Methods in Natural Language Processing. (**EMNLP'23**)
- [40] J. Li, W. Gao, Q. Lei, and **D. Xu**. Breaking through Deterministic Barriers: Randomized Pruning Mask Generation and Selection [C]. The 2023 Conference on Empirical Methods in Natural Language Processing. (**EMNLP'23, Findings**)
- [41] J. Gu, Z. Nan, Z. Peng, X. Shen, and **D. Xu**. Co-evolving Data-driven and NLU-driven Synthesizers for Generating Code in Domain Growth and Data Scarcity [C]. The 2023 Conference on Empirical Methods in Natural Language Processing. (**EMNLP'23, Pan-DL Workshop**)
- [42] D. Zhu, B. Lei, J. Zhang, Y. Fang, Y. Xie, R. Zhang, and **D. Xu**. Rethinking Data Distillation: Do Not Overlook Calibration [C]. International Conference on Computer Vision (**ICCV'23**)
- [43] J. Wang, X. Yang, S. Cui, L. Che, L. Lyu, **D. Xu**, and F. Ma. Towards Personalized Federated Learning via Heterogeneous Model Reassembly [C]. The 37th Annual Conference on Neural Information Processing Systems (**NeurIPS'23**)
- [44] S. Li, H. Mei, J. Li, H. Wei, and **D. Xu**. Toward Efficient Traffic Signal Control: Smaller Network Can Do More [C]. 62nd IEEE Conference on Decision and Control (**CDC'23**)
- [45] B. Lei, R. Zhang, **D. Xu**, and B. K. Mallick. Calibrating the Rigged Lottery: Making All Tickets Reliable [C]. The 11th International Conference on Learning Representations (**ICLR'23**)
- [46] Q. Zhang, S. Chen, **D. Xu**, Q. Cao, X. Chen, T. Cohn, and M. Fang. A Survey for Efficient Open Domain Question Answering [C]. The 61th Annual Meeting of the Association for Computational Linguistics (**ACL'23**)

- [47] L. Wu, B. Lei, **D. Xu**, and D. Zhou. Towards Reliable Rare Category Analysis on Graphs via Individual Calibration [C]. The 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD'23**)
- [48] C. Liu, D. Doshi, M. Bhargava, R. Shang, J. Cui, **D. Xu**, and E. Gehringer. Labels Are Not Necessary: Assessing Peer-Review Helpfulness Using Domain Adaptation Based on Self-Training [C]. The 18th Workshop on Innovative Use of NLP for Building Educational Applications (**BEA'23**)
- [49] Z. Dong, **D. Xu**. Exploring the Augmented Large Language Model with Mathematical tools in Personalized and Efficient Education [C]. The 6th International Conference on Artificial Intelligence and Big Data (**ICAIBD'23**)
- [50] Y. Liu, M. Chen, **D. Xu**, Z. Yang, and S. Zhao. E-App: An Environment-Aware Access Point Planning Framework for mmWave Wireless LANs [C]. The 32nd International Conference on Computer Communications and Networks (**ICCCN'23, Best Paper Award**)
- [51] Y. X., D. Zhu, B. Lei, **D. Xu**, and R. Zhang. Efficient Informed Proposals for Discrete Distributions via Newton's Series Approximation [C]. The 26th International Conference on Artificial Intelligence and Statistics (**AISTATS'23**)
- [52] S. Huang, H. Fang, K. Mahmood, B. Lei, N. Xu, B. Lei, Y. Sun, **D. Xu**, Wu. Wen, and C. Ding. Neurogenesis Dynamics-inspired Spiking Neural Network Training Acceleration [C]. The 60th Design Automation Conference (**DAC'23**)
- [53] S. Huang, B. Lei, **D. Xu**, H. Peng, Y. Sun, M. Xie, and C. Ding. Dynamic Sparse Training via Balancing the Exploration-Exploitation Trade-off [C]. The 60th Design Automation Conference (**DAC'23**)
- [54] J. Li, T. Zhang, E. Yan, and **D. Xu**. FP8-BERT: Post-Training Quantization for Transformer [C]. The 1st Workshop on DL-Hardware Co-Design for AI Acceleration (**DCAA'23**)
- [55] Y. Xie, Z. Li, H. Bao, X. Jia, **D. Xu**, X. Zhou, and S. Skakun. Auto-CAM: Label-Free Earth Observation Imagery Composition and Masking Using Spatio-Temporal Dynamics [C]. The 37th AAAI International Conference on Artificial Intelligence (**AAAI'23**)
- [56] D. Luo, W. Cheng, Y. Wang, **D. Xu**, J. Ni, W. Yu, X. Zhang, Y. Liu, Y. Chen, H. Chen, and X. Zhang. Time Series Contrastive Learning with Information-Aware Augmentations [C]. The 37th AAAI International Conference on Artificial Intelligence (**AAAI'23**)
- [57] Y. Tian, W. Gao, Q. Zhang, P. Sun, and **D. Xu**. Improving long-tailed classification by disentangled variance transfer [J]. **Internet of Things (2023)**: 100687.
- [58] **D. Xu**, S. Mukherjee, X. Liu, D. Dey, W. Wang, X. Zhang, A. H. Awadallah, and J. Gao. Few-shot Task-agnostic Neural Architecture Search for Distilling Large Language Models [C]. The 36th Annual Conference on Neural Information Processing Systems (**NeurIPS'22**)
- [59] I. Yen, Z. Xiao, and **D. Xu**. S4: a High-sparsity, High-performance AI Accelerator [C]. Sparsity in Neural Networks 2022 Workshop (**SNN'22**)
- [60] S. Huang, N. Liu, Y. Liang, H. Peng, H. Li, **D. Xu**, M. Xie, and C. Ding. An Automatic and Efficient BERT Pruning for Edge AI Systems [C]. The 23rd IEEE International Society for Quality Electronic Design (**ISQED'22**)

- [61] S. Huang*, **D. Xu***, I. Yen, S. Chang, B. Li, S. Chen, M. Xie, H. Liu, and C. Ding. Sparse Progressive Distillation: Resolving Overfitting under Pretrain-and-Finetune Paradigm [C]. The 60th Annual Meeting of the Association for Computational Linguistics (**ACL'22**)
Acceptance rate: 714/3350=21.3%
- [62] **D. Xu**, W. Cheng, D. Luo, H. Chen, and X. Zhang. InfoGCL: Information-Aware Graph Contrastive Learning [C]. The 35th Annual Conference on Neural Information Processing Systems (**NeurIPS'21**)
Acceptance rate: 2372/9122=26.0%
- [63] **D. Xu**, I. Yen, J. Zhao, and Z. Xiao. Rethinking Network Pruning - under the Pre-train and Fine-tune Paradigm [C]. 2021 Annual Conference of the North American Chapter of the Association for Computational Linguistics (**NAACL-HLT'21**)
Acceptance rate: 477/1797=26.5%
- [64] X. Dong, Y. Zhu, Z. Fu, **D. Xu**, and G. de Melo. Data Augmentation with Adversarial Training for Cross-Lingual NLI [C]. The 59th Annual Meeting of the Association for Computational Linguistics (**ACL'21**)
Acceptance rate: 714/3350=21.3%
- [65] **D. Xu**, W. Cheng, J. Ni, D. Luo, Masanao Natsumeda, D. Song, B. Zong, H. Chen, and X. Zhang. Deep Multi-Instance Contrastive Learning with Dual Attention for Anomaly Precursor Detection [C]. The 21th SIAM International Conference on Data Mining (**SDM'21**)
Acceptance rate: 85/400=21.3%
- [66] **D. Xu**, W. Cheng, X. Dong, B. Zong, W. Yu, J. Ni, D. Song, X. Zhang, H. Cheng, and X. Zhang. Multi-Task Recurrent Modular Networks [C]. The 35th AAAI International Conference on Artificial Intelligence (**AAAI'21**)
Acceptance rate: 1692/7911=21.4%
- [67] **D. Xu**, J. Liang, W. Cheng, H. Wei, H. Cheng, and X. Zhang. Transformer Style Relational Reasoning with Dynamic Memory Updating for Temporal Network Modeling [C]. The 35th AAAI International Conference on Artificial Intelligence (**AAAI'21**)
Acceptance rate: 1692/7911=21.4%
- [68] H. Wei, **D. Xu**, J. Liang, and Z. Li. How Do We Move: Modeling Human Movement with System Dynamics [C]. The 35th AAAI International Conference on Artificial Intelligence (**AAAI'21**)
Acceptance rate: 1692/7911=21.4%
- [69] J. Liang, Y. Wu, **D. Xu**, and V. Honavar. Longitudinal Deep Kernel Gaussian Process Regression [C]. The 35th AAAI International Conference on Artificial Intelligence (**AAAI'21**)
Acceptance rate: 1692/7911=21.4%
- [70] D. Luo, W. Cheng, **D. Xu**, W. Yu, B. Zong, H. Chen, and X. Zhang. Parameterized Explainer for Graph Neural Network [C]. The 34th Annual Conference on Neural Information Processing Systems (**NeurIPS'20**)
Acceptance rate: 1900/9454=20.1%
- [71] X. Dong, Y. Zhu, Y. Zhang, Z. Fu, **D. Xu**, S. Yang, and G. de Melo. Leveraging Adversarial Training in Self-Learning for Cross-Lingual Text Classification [C]. The 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval (**SIGIR'20**)
Acceptance rate: 300/1062=28.2%

- [72] **D. Xu**, W. Cheng, B. Zong, D. Song, J. Ni, W. Yu, Y. Liu, H. Chen, and X. Zhang. Tensorized LSTM with Adaptive Shared Memory for Learning Trends in Multivariate Time Series [C]. The 34th AAAI International Conference on Artificial Intelligence (**AAAI'20**)
Acceptance rate: $1591/7737=20.6\%$
- [73] J. Liang, **D. Xu**, Y. Sun, and V. Honavar. Longitudinal Multi-Level Factorization Machines [C]. The 34th AAAI International Conference on Artificial Intelligence (**AAAI'20**)
Acceptance rate: $1591/7737=20.6\%$
- [74] **D. Xu**, W. Cheng, D. Luo, X. Liu, and X. Zhang. Spatio-Temporal Attentive RNN for Node Classification in Temporal Attributed Graphs [C]. The 28th International Joint Conference on Artificial Intelligence (**IJCAI'19**)
Acceptance rate: $850/4752=17.9\%$
- [75] **D. Xu**, W. Cheng, D. Luo, Yameng Gu, X. Liu, J. Ni, B. Zong, H. Chen, and X. Zhang. Adaptive Neural Network for Node Classification in Dynamic Networks [C]. The 19th IEEE International Conference on Data Mining (**ICDM'19**)
Acceptance rate: $183/930=19.7\%$
- [76] **D. Xu**, W. Cheng, B. Zong, J. Ni, D. Song, W. Yu, Y. Chen, H. Chen, and X. Zhang. Deep Co-Clustering [C]. The 19th SIAM International Conference on Data Mining (**SDM'19**)
Acceptance rate: $90/397=22.7\%$
- [77] J. Ni, S. Chang, X. Liu, W. Cheng, H. Chen, **D. Xu**, and X. Zhang. Co-Regularized Deep Multi-Network Embedding [C]. The 27th International Conference on World Wide Web (**WWW'18**)
Acceptance rate: $170/1175=14.5\%$
- [78] Y. Tian, **D. Xu**, and C. Zhang. A Review of Multi-Instance Learning Research [J]. Operations Research Transactions, 2018, 02: 1-17
- [79] **D. Xu**, J. Wu, D. Li, Y. Tian, X. Zhu, and X. Wu. SALE: Self-Adaptive LSH Encoding for Multi-Instance Learning [J]. **Pattern Recognition**, 2017 (**7.74 impact factor**)
- [80] D. Li, **D. Xu**, J. Tang, and Y. Tian. Metric Learning for Multi-Instance Classification with Collapsed Bags [C]. The 30th IEEE International Joint Conference on Neural Networks (**IJCNN'17**)
- [81] D. Li, W. Zhang, **D. Xu**, and Y. Tian. Multi-Metrics Classification Machine [C]. International Conference on Information Technology and Quantitative Management (**ITQM'16**)
- [82] **D. Xu**, and Y. Tian. A Comprehensive Survey of Clustering Algorithms [J]. Annals of Data Science, 2015, 2(2): 165-193
- [83] **D. Xu**, T. Chen, and W. Xu. A Support Vector Machine-Based Ensemble Prediction for Crude Oil Price with VECM and STEPMRS [J]. International Journal of Global Energy Issues, 2015
- [84] **D. Xu**, Y. Zhang, C. Cheng, W. Xu, and L. Zhang. A Neural Network-Based Ensemble Prediction Using PMRS and ECM [C]. The 47th IEEE Hawaii International Conference on System Sciences (**HICSS'14**)

INDUSTRY
EXPERIENCE

Microsoft Research (MSR), Redmond, WA

2021

- Research Intern, Mentors: **Subho Mukherjee**, **X. Liu**, **D. Dey**, **A. H. Awadallah**, **J. Gao**
- Project: Task-agnostic Auto-Transformer Search [NeurIPS 2022]

	Moffett.AI , Los Altos, CA 2020 - Research Intern, Mentor: I. Yen, Co-founder - Project: Data-free Model Compression [NAACL 2021 & a U.S. patent]
	NEC Labs America , Princeton, NJ 2019 - Research Intern, Mentor: W. Cheng - Project: Knowledge Transfer in Multi-Task Learning [AAAI 2021] - Project: Trend Learning in Multivariate Time Series [AAAI 2020]
	NEC Labs America , Princeton, NJ 2018 - Research Intern, Mentor: W. Cheng, Senior Researcher - Project: Contrastive Anomaly Detection [SDM 2021]
ACADEMIA EXPERIENCE	Penn State University 2017-2022 - Graduate Research Assistant, Adviser: X. Zhang - Thesis: Resource-efficient Deep Learning: Democratizing AI at Scale
	Chinese Academy of Sciences , Beijing, China 2014-2017 - Graduate Research Assistant, Adviser: Y. Tian - Thesis: Efficient Multi-instance Learning
	Renmin University of China , Beijing, China 2012-2014 - Undergraduate Research Assistant, Adviser: W. Xu - Thesis: Ensemble Forecasting Model for Time Series Data
TEACHING EXPERIENCE	Instructor at NC State - CSC 422: Automated Learning and Data Analysis Spring'23, Fall'24, 25, 26 Course Materials: Introduction to Data Mining (Second Edition) - CSC 791&591: Advanced Topics in Efficient Deep Learning Fall'22, 23, Spring'24, 25, 26 Course Materials: Dive into Deep Learning
	Teaching Assistant at Penn State - SRA 268, Visual Analytics Fall 2021 Instructor: Prof. Mahir Akgun Course Materials: Visual Analytics with Tableau - SRA 450, Cybercrime and Cyberwar Fall 2021 Instructor: Prof. John Hodgson Course Materials: Cybersecurity: What Everyone Needs to Know - DS/CMPSC 410, Programming Models for Big Data Spring 2021 Instructor: Prof. John Yen Course Materials: Learning Spark - SRA 365, Statistics for Security and Risk Analysis Fall 2020 Instructor: Dr. James Farrugia Course Materials: Discovering Statistics Using R - DS 402, Introduction to Social Media Mining Spring 2020 Instructor: Prof. Suhang Wang Course Materials: Social Media Mining: An Introduction - SRA 365, Statistics for Security and Risk Analysis Spring 2019 Instructor: Dr. Katherine Hamilton Course Materials: Foundations and Practice of Intermediate Statistics

- IST 210, Organization of Data Fall 2018
Instructor: Prof. [X. Zhang](#)
Course Materials: [Database Systems Concepts](#)
(The Award for Excellence in Teaching Support)

Guest Lecturer

- COSI 133A, Graph Mining Fall 2021
Brandeis University, Slides [\[Link\]](#)
- COSI 165B, Deep Learning Spring 2021
Brandeis University, Slides: [\[Link\]](#)

MENTORING EXPERIENCE

Ph.D. Students

- Muhammad Alahmadi
Research Interest: Improving Data-Centric Reliability for Scalable AI Systems
M.S.: North Carolina State University, USA
B.S.: King Fahd University of Petroleum and Minerals (Second Honors), Saudi Arabia
- Aditya Basarkar
Research Interest: LLM-Powered Complex Mathematical Reasoning
B.S.: North Carolina State University (Dean's List, GPA: 3.974), USA
Majors: B.S. in Computer Science, B.S. in Statistics
- Li-Chia (Jerry) Chang
Research Interest: Engineering Efficient Embodied AI for Adaptive Intelligent Systems
M.S.: North Carolina State University, USA
B.S.: National Chi Nan University, Taiwan
- Bowen (Berwin) Chen (co-advised with Drs. Roy He and Paul Liu)
Research Interest: Advancing Generative Models for Environmental Intelligence
M.S.: University of Birmingham, United Kingdom
B.S.: Hohai University, China
- Vignes KV
Research Interest: Enhancing Generative Models for Super-Resolution Image Denoising
M.S.: North Carolina State University, USA
B.S.: North Carolina State University (University Honors Program Scholar), USA
- Stephen Liu
Research Interest: Advancing Efficient AI Models for Biomedical and Genomic Discovery
M.S.: North Carolina State University, USA
B.S.: University of Michigan, USA
- Travis Thompson
Research Interest: Advancing Reliable Foundation Models for Scientific Discovery
B.S.: Auburn University (Undergraduate Research Fellow), USA
- Benjamin T. Tabarsi (co-advised with Dr. Tiffany Barnes)
Research Interest: Developing Trustworthy AI Agents for Personalized Learning
M.S.: Science and Research Branch of Azad University, Iran
B.S.: University of Mazanadran, Iran

Undergraduate Students

- Precious Donkor
Research Interest: Exploring and Mitigating Implicit Bias in Large Language Models
B.S.: North Carolina State University (GPA: 3.9), USA
Achievements: i) Federal Work-Study Award, ii) NC State COE REU Award
Publication: AAAI 2025 Undergraduate Consortium

- Ethan Xu
Research Interest: Developing Multimodal AI Agents for Child–Robot Learning and Instruction Grounding
B.S.: North Carolina State University, USA
Achievements: Two NC State COE REU Awards (Summer and Fall 2026)

Master Students

- Nitya Naga Sai Atluri
Research Interest: Unveiling Implicit Bias in LLM-Generated Educational Materials
B.S.: Koneru Lakshmaiah Education Foundation - Andhra Pradesh, India
- Xinyuan (Teddy) Chen
Research Interest: Trustworthiness Evaluation of LLMs
B.S.: Fudan University, China
- Harsh Mauny
Research Interest: AI & Humans Co-Design Artistic Creations
B.S.: Sardar Vallabhbhai Patel Institute of Technology, India
- Homak Patel
Research Interest: Reliable Retrieval-Augmented Generation for Education
B.S.: Ahmedabad University, Gujarat, India

Alumni

- Zhiyuan Peng, Postdoc, 2023-2024, now Meta
Topic: Tool-Augmented Large Language Model
- Shengkun Tang, PhD, 2023-2023, now MBZUAI
Topic: Multi-Modal Foundation Models
- Chengyuan Liu, PhD, 2022-2024
Topic: Data-Efficient Large Language Models for Education
- Zihan (Z) Dong, Undergraduate, 2023-2024, now Georgia Tech
Topic: Reliable Large Language Models for Education
- John Zhu, Master, 2024, now NC State
Topic: Trustworthy Evaluation of LLMs
- Bowen Lei, Intern, 2022-2024, now Apple
Topic: Theoretical Foundations of Efficient Learning
- Zhengdong Zhang, Intern, 2023-2024, now Amazon
Topic: Reliable Large Language Models for Education
- Binfeng Xu, Intern, 2023-2023, now Samsung
Topic: Tool-Augmented Large Language Model

PROFESSIONAL
SERVICE

Panelists

- NSF Programs, 2023-2026

Column Editor

- ACM SIGAI Newsletter

Workshop Chair & Co-Chair

- The 3rd Workshop on DL-Hardware Co-Design for Generative AI Acceleration @DAC2026
- The 2nd Workshop on DL-Hardware Co-Design for Generative AI Acceleration @DAC2025
- The 4th Workshop on Resource-Efficient Learning for Knowledge Discovery @KDD2026
- The 3rd Workshop on Resource-Efficient Learning for Knowledge Discovery @KDD2025
- The 1st Workshop on DL Co-Design for Generative AI Acceleration @DAC2024

- The First Workshop on Dataset Distillation for Computer Vision @CVPR2024
- The 2nd Workshop on Resource-Efficient Learning for Knowledge Discovery @KDD2024
- The 1st Workshop on DL-Hardware Co-Design for AI Acceleration @AAAI2023
- The 1st Workshop on Resource-Efficient Learning for Knowledge Discovery @KDD2023

Session Chair

- Scalable, Distributed Systems & Trustable AI @KDD2022
- Deep Learning: New Architectures and Models @KDD2022

Area Chair

- The Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING) 2024

Senior Program Committee

- AAAI Conference on Artificial Intelligence (AAAI) 2024, 2025, 2026
- International Joint Conferences on Artificial Intelligence (IJCAI) 2021

Program Committee

- Neural Information Processing Systems (NeurIPS) 2020, 2021, 2022, 2023, 2024, 2025
- International Conference on Learning Representations (ICLR) 2021, 2022, 2023, 2024
- International Conference on Machine Learning (ICML) 2021, 2022, 2023
- AAAI Conference on Artificial Intelligence (AAAI) 2020, 2021, 2022, 2023
- SIGKDD Conference on Knowledge Discovery and Data Mining (KDD) 2020-2023
- Association for Computational Linguistics (ACL) Rolling Review 2022
- North American Chapter of the Association for Computational Linguistics (NAACL) 2021
- Conference on Empirical Methods in Natural Language Processing (EMNLP) 2020, 2021
- International Conference on Computational Linguistics (COLING) 2022
- Learning on Graphs Conference (LoG) 2022
- International Joint Conferences on Artificial Intelligence (IJCAI) 2020, 2022
- ACM International Conference on Web Search and Data Mining (WSDM) 2022
- SIAM International Conference on Data Mining (SDM) 2022
- European Chapter of the Association for Computational Linguistics (EACL) 2021
- Conference on Information and Knowledge Management (CIKM) 2020, 2021, 2022
- Asia-Pacific Chapter of the Association for Computational Linguistics & International Joint Conference on Natural Language Processing (AACL-IJCNLP) 2020
- International Joint Conference on Neural Networks (IJCNN) 2018, 2019, 2020, 2021

Journal Reviewer

- IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)
- IEEE Transactions on Neural Networks and Learning Systems (TNNLS)
- IEEE Transactions on Knowledge and Data Engineering (TKDE)
- IEEE Transactions on Cybernetics
- Information Fusion
- ACM Transactions on Knowledge Discovery from Data (TKDD)
- Pattern Recognition
- Neural Networks
- ACM Transactions on Asian and Low-Resource Language Information Processing
- IEEE Access
- Neural Computation
- Complexity
- Soft Computing
- Journal of Sports Engineering and Technology
- Complex & Intelligent Systems
- Multimedia Tools and Applications
- Big Data

Conference Volunteer

- The Annual Conference of NAACL-HLT, 2021
- Backuping SDM Session Chairs, 2021
- The 35th AAAI Conference on Artificial Intelligence, 2021
- The 26th SIGKDD Conference on Knowledge Discovery and Data Mining, 2020

PATENTS

Issued U.S. Patents

- Tensorized LSTM with Adaptive Shared Memory for Learning Trends in Multivariate Time Series
W. Cheng, H. Chen, J. Ni, **D. Xu**, and W. Yu
U.S. Patent Nos. [11,783,181](#) (Oct. 10, 2023) and [12,579,436](#) (Mar. 17, 2026)
- Information-Aware Graph Contrastive Learning
W. Cheng, **D. Xu**, and H. Chen
U.S. Patent Nos. [12,488,238](#) (Dec. 2, 2025); [12,511,540](#) (Dec. 30, 2025); [12,518,162](#) (Jan. 6, 2026); and [12,518,163](#) (Jan. 6, 2026)
- Neural Network Pruning Method and System via Layerwise Analysis
E. Yan, **D. Xu**, and J. Liu
U.S. Patent No. [12,340,313](#) (Jun. 24, 2025)
- Bank-Balanced-Sparse Activation Feature Maps for Neural Network Models
E. Yan, **D. Xu**, and J. Liu
U.S. Patent No. [12,236,341](#) (Feb. 25, 2025)
- Spatio-Temporal Gated Recurrent Unit
W. Cheng, H. Chen, and **D. Xu**
U.S. Patent No. [11,461,619](#) (Oct. 4, 2022)
- System and Method for Knowledge-Preserving Neural Network Pruning
E. Yan, **D. Xu**, and Z. Xiao
U.S. Patent No. [11,200,497](#) (Dec. 14, 2021)

Pending U.S. Patent Application

- Logits-Based Detector Without Logits from Black-Box LLMs
W. Cheng, H. Chen, **D. Xu**, and X. Yang
U.S. Patent Application No. 19/194,706; Publication No. [US20250356126A1](#) (filed Apr. 30, 2025; pending)

PATENTS

- Spatio Temporal Gated Recurrent Unit
W. Cheng, H. Chen, and **D. Xu**
U.S. Patent. [11,461,619](#). Oct. 2022
- System and Method for Knowledge-Preserving Neural Network Pruning
E. Yan, **D. Xu**, and Z. Xiao
U.S. Patent. [11,200,497](#). Dec. 2021

PROFESSIONAL TALKS

- Scalable, Efficient, and Grounded Agentic AI for Engineering Workflows
NCSU COE Applied AI Summer Workshop, July 2026
- From AI to Agentic Systems: Enabling Intelligent Decision-Making in Food Science
NCSU Food Science Department IPAC Meeting, April 2026
- Deep Learning Basics and AI Agents
NCSU Food Science Department AI Forum, March 2026

- Building Scalable and Efficient Agentic AI Systems
SAS Institute, Cary, NC, USA, April 2025
- Building Scalable and Efficient Agentic AI Systems
NCSU MSE Department, Raleigh, NC, USA, April 2025
- Advancing Real-World Edge Intelligence
ABB Corporate Research, Raleigh, NC, USA, Jan. 2025
- Agentic LLM-Powered Diagnostic Chatbot for SAP Manufacturing
ABB Corporate Research, Raleigh, NC, USA, Sep. 2024
- The Impact of AI on Our Lives and Beyond
Fo Guang Shan Buddhist Temple, North Carolina, March 2024
- Leveraging Foundation Models for Enhanced Geospatial Analytics and Conservation
Forest Carbon Solutions Initiative (FCSI), NC State, March 2024
- How LLMs Work and Cutting-Edge Research on Generative AI
STARS AI Scholars Program, Dec 2023 [remote]
- Sculpting the Future of Collective Growth in Collaborative AI
Microsoft Research Asia, Beijing, China, Sep 2023 [remote]
- ChatGPT in Corporate Real Estate - Unlocking the Potential [[link](#)]
CoreNet Global, Raleigh, NC, USA, Aug 2023
- Testing Accuracy is Not All You Need: Less Training Cost & More Testing Reliability
Rutgers University, New Brunswick, USA, Feb 2023
- Resource-efficient Deep Learning: Democratizing AI at Scale
Pinterest, San Francisco, USA, Aug 2022
- Resource-efficient Deep Learning: Democratizing AI at Scale
Amazon Search (A9), USA, May 2022
- Resource-efficient Deep Learning: Democratizing AI at Scale
Vanderbilt University, Nashville, USA, April 2023
- Resource-efficient Deep Learning: Democratizing AI at Scale
University of Connecticut, Stamford, USA, April 2023
- Parameter Efficiency: Democratizing AI at Scale [[Slides](#)]
Brandeis University, Waltham, USA, Dec 2021
- Chasing Efficiency of Pre-trained Language Models
Microsoft Research Lab, Redmond, Washington, USA, Jun 2021
- BERT Pruning: Structural vs. Sparse [[Slides](#)]
Brandeis University, Waltham, USA, Apr 2021
- BERT, Compression, and Applications [[Slides](#)]
Xpeng Motors, Mountain View, USA, Apr 2021
- BERT Architecture and Computation Analysis
Moffett.AI, Los Altos, USA, May 2020.
- Anomaly Precursor Detection via Multi-Instance Contrastive Learning
NEC Laboratories America, Princeton, USA, May 2019
- Efficient Multiple Instance Learning [[Slides](#)]
NEC Laboratories America, Princeton, USA, May 2018

HONORS,
AWARDS, AND
DISTINCTIONS

Faculty Honors and Research Recognition

- NC State College of Engineering Dean's Applied AI Research Accelerator Award 2026
- AAAI Best Technical Demonstration Award 2025
- AAAI New Faculty Highlights 2025
- Best Poster Award, AAAI Spring Symposium on GenAI@Edge 2025
- NVIDIA Academic Grant Program Award 2025
- Microsoft Accelerating Foundation Models Research Award 2024
- NC State Carla Savage Award 2024
- Best Paper Award Runner-Up, IEEE IPCCC 2024
- Best Paper Award, IEEE ICCCN 2023

Selected Publication Distinctions

- Oral Presentation, ICLR (top 5%) 2026
- Oral Presentation, AAAI (top 5%) 2025
- Cover Article, Analytical Chemistry 2025
- Highlight Paper, CVPR (top 2.5%) 2023

The Pennsylvania State University

- Third Place Winner (Engineering), 37th Annual Graduate Exhibition 2022
- NAACL Scholarship 2021
- SIAM Student Travel Award 2021
- KDD Student Registration Award 2020
- AAAI Student Scholarship 2020
- College of IST Award for Excellence in Teaching Support (Top 2) 2019
- IST Travel Award 2019–2021

University of Chinese Academy of Sciences

- Chinese Academy of Sciences President's Fellowship 2016
- National Graduate Scholarship, China (top 2% at the university) 2016
- Graduate Student Academic Scholarship 2015–2017

Renmin University of China

- First-Class Scholarship, Sashixuan Elite Fund (top 5% at the university) 2014
- Kwang-hua Scholarship 2014
- Second-Class Scholarship for Excellent Student Cadre 2014
- Meritorious Winner, Mathematical Contest in Modeling 2013
- First-Class Scholarship for Social Work and Volunteer Service 2013

PROFESSIONAL
MEMBERSHIPS

- Institute of Electrical and Electronics Engineers (IEEE) 2023–Present
- Association for Computing Machinery (ACM) 2025–Present
- Association for the Advancement of Artificial Intelligence (AAAI) 2025–Present